

Myopic Information-Gain Selection over a Theory Ensemble

Setup

At a selection point we generate an ensemble of M theories $T_1, \dots, T_M$, ranked 1 (most likely) to M . The candidate question set is the unanswered bank $\mathcal{Q}$ (after it has been augmented with the crux questions).

1 Prior over theories (rank $\rightarrow$ weight)

`assign_rank_weights` puts a geometric prior on rank, with decay $\gamma \in (0, 1]$:

$$\pi_k = P(T = T_k) = \frac{\gamma^{r_k-1}}{\sum_{j=1}^M \gamma^{r_j-1}}, \quad r_k = \text{rank of } T_k.$$

$\gamma < 1$ concentrates mass on top-ranked theories; $\gamma = 1$ gives a uniform prior. (Default $\gamma = 0.6$.)

2 Per-theory answer predictions

For each candidate question q_i and theory T_k , one `_predict_answer_vector` call yields a label in $\{\text{YES}, \text{NO}, \text{UNKNOWN}\}$, mapped to the theory's predicted probability that the answer is YES:

$$a_{k,i} = P(A_i = \text{YES} \mid T_k) = \begin{cases} 1 & \text{YES} \\ 0 & \text{NO} \\ \frac{1}{2} & \text{UNKNOWN (agnostic)} \end{cases}$$

So the per-theory binary answer model is $P(A_i = \text{YES} \mid T_k) = a_{k,i}$, $P(A_i = \text{NO} \mid T_k) = 1 - a_{k,i}$. Cost: M LLM calls total (independent of $|\mathcal{Q}|$), versus B-diff's $\approx 4|\mathcal{Q}|$.

3 Marginal answer and Bayesian posterior

Marginal probability of a YES under the prior:

$$\rho_i = P(A_i = \text{YES}) = \sum_{k=1}^M \pi_k a_{k,i}.$$

Posterior over theories after observing each answer (Bayes):

$$P(T_k \mid A_i = \text{YES}) = \frac{\pi_k a_{k,i}}{\rho_i}, \quad P(T_k \mid A_i = \text{NO}) = \frac{\pi_k (1 - a_{k,i})}{1 - \rho_i}.$$

4 Score = mutual information (expected info gain over theory identity)

$$\text{score}(q_i) = I(A_i; T) = H(T) - \mathbb{E}_{A_i}[H(T | A_i)]$$

with everything in bits:

$$\begin{aligned} H(T) &= - \sum_k \pi_k \log_2 \pi_k, \\ H(T | A_i) &= \rho_i H(T | A_i=\text{YES}) + (1 - \rho_i) H(T | A_i=\text{NO}), \\ H(T | A_i=a) &= - \sum_k P(T_k | A_i=a) \log_2 P(T_k | A_i=a). \end{aligned}$$

This is exactly `mutual_information(weights, p_yes_per_theory)`: it normalizes `yes_weights =  $\pi_k a_{k,i}$` and `no_weights =  $\pi_k (1 - a_{k,i})$` to get the two posteriors, weights their entropies by ρ_i , $1 - \rho_i$, and returns `max(0,  $H(T) - H(T | A_i)$ )`.

Equivalent dual form (cheaper to state, useful as a sanity check — the code uses the form above):

$$I(A_i; T) = H(A_i) - H(A_i | T) = H_b(\rho_i) - \sum_k \pi_k H_b(a_{k,i}),$$

where $H_b(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$ is the binary entropy. From this it is immediate that:

- theories agree ($a_{k,i}$ all equal) $\Rightarrow H(A_i | T) = H_b(\rho_i) \Rightarrow I = 0$ (`test_mi_zero_when_theories_agree`);
- two equal-weight theories that disagree ($a = \{1, 0\}$, $\pi = \{\frac{1}{2}, \frac{1}{2}\}$) $\Rightarrow I = H_b(\frac{1}{2}) = 1$ bit (`test_mi_max_for_even_split`);
- an all-UNKNOWN question scores 0 (each $H_b(\frac{1}{2})$ cancels).

5 Selection

Probe-selection (the LLM) narrows the bank to a candidate set S ; the pursued target is

$$q^* = \arg \max_{q_i \in S} I(A_i; T),$$

with an LLM tie-break if the top score is shared. The experiment plan is then formulated from q^* . The crux questions are, by construction, drafted to make $a_{\cdot, i}$ disagree across theories — i.e. to be high- $I(A_i; T)$ candidates injected directly into $\mathcal{Q}$.

What this is not (the B vs. A boundary)

This is the myopic, one-step, stateless version. The prior π comes only from the LLM’s rank ordering, and T is regenerated fresh each selection point — there is no carried-forward posterior update from the actually observed outcome:

$$\pi_k \not\leftarrow \frac{\pi_k P(o_t | T_k)}{\sum_j \pi_j P(o_t | T_j)}.$$

Adding that recursive Bayesian update (theories persist across steps, weights updated by observed transitions, action chosen to maximize I over a horizon) is precisely the step up to Plan A.